HISAABSET (LedgerLoop)

B2B Receivables Decision Desk & Automated Collections Engine

FyndHacks 2026 — Track C28 Submission Package

Submission Track:	C28 Receivables Follow-Up Desk & Retailer Credit Decision Engine
Submitting Lead:	Muhammad Khaja Zeeshan Ali (B.Tech) — zeeshanalimuhammadkhaja@gmail.com
Team Members:	Faiz Sayeed, Sai Kumar Reddy
Exact Commit SHA:	1947a631125956e2f5db3c3d51b21232da984bfb
GitHub Repository:	https://github.com/MdZeeshanAli01/LedgerLoop
Live Cloud URL:	http://13.201.115.214:8501 (AWS ECS Fargate ap-south-1 Mumbai)
Demo Credentials:	Role: Finance Lead | User: **admin** | Password: **hisaabset123**

1. Executive Summary & Problem Statement

In India, over 350,000 organized B2B wholesale distributors (FMCG, pharmaceuticals, hardware, lubricants) operate under intense working capital strain. While accounting tools like Tally and Busy record invoices, they do not resolve the primary cash-leakage bottleneck: **trapped receivables and disputed collections**.

- **Working Capital Trap:** Distributors routinely carry ■10L–■25L in 30–90+ day overdue credits, paying 12%–14% p.a. in bank cash-credit/overdraft interest (■10,000–■18,000/month in dead financing costs).
- **The Dispute Hostage Problem:** Retailers exploit minor delivery discrepancies (e.g., a ■2,000 damaged jar on a ■50,000 bill) to withhold 100% of the invoice payment for months.
- **Follow-up Paralysis:** Traditional collections rely on clumsy WhatsApp messages or aggressive phone calls that strain long-term distributor-retailer commercial goodwill.

2. The Hisaabset Working Solution

Hisaabset is an intelligent receivables decision-support desk designed for the distributor Finance Lead. Key operational capabilities include:

- **Dispute Isolation Engine:** Instantly separates disputed claims (damaged goods, rate differences, short deliveries) from safe collectible cash. The undisputed ■48,000 is collected immediately while the ■2,000 discrepancy undergoes formal credit-note reconciliation.
- **Deterministic Priority Scoring:** Ranks debtors mathematically using a formula combining Age Score (0.45), Relative Balance Score (0.35), and Broken-Promise History (0.20), plus non-linear escalation for debts >90 days.
- **AI Ingestion & OCR:** Ingests PDF and physical invoices directly using Google Gemini 1.5 Flash, automatically extracting line items, dates, and amounts without manual keying.
- **One-Click Payment Settlement & Vouchers:** Records incoming RTGS/NEFT/Cash payments, marks invoices as Closed/Paid, and generates printable Payment Settlement Vouchers.
- **Human-in-the-Loop Reminders:** Previews tone-adapted (Gentle, Firm, Formal) collection notices dispatched securely via Amazon SES with verified audit trails.

3. Cloud-Native Production Architecture (AWS)

Layer	AWS Service	Specification & Role
Compute Runtime	AWS ECS on Fargate	0.5 vCPU, 1 GB RAM continuous container microservice (ap-south-1 Mumbai)
Container Registry	AWS ECR	Immutable Docker image repository with automated digest scanning
Communications	Amazon SES	Managed enterprise transactional email engine with verified DKIM/SPF

Networking & Security	VPC Security Groups & IAM	Port 8501 restricted ingress; least-privilege IAM execution roles; CloudWatch logging
Testing & Quality	Automated Pytest Suite	89 unit and integration tests passing; zero Decimal money precision loss

4. Business Model, AWS TCO & Unit Economics

- **SaaS Subscription Pricing:** ■2,499 / month per distributor (or ■24,990 / year upfront).
- **AWS Cloud TCO:** $22.16 / month (~■1,840 / month) for 24/7 continuous hosting on AWS ECS Fargate.
- **Multi-Tenant Scalability:** A single 0.5 vCPU base pod serves 30–50 SMB distributors, reducing infrastructure cost to <■65 per tenant/month (>97% gross margin).
- **Distributor ROI:** By accelerating payment cycles by 12–15 days and preventing ■15,000+ in monthly overdraft interest, Hisaabset delivers an undeniable **6x–8x net ROI**.

5. Key Repository Evidence Links

- **Live Application:** http://13.201.115.214:8501
- **GitHub Repository:** https://github.com/MdZeeshanAli01/LedgerLoop
- **Financial & Cloud Cost Model:** docs/BUSINESS_PRICING_AND_AWS_COST_MODEL.md
- **Technical Handoff & Architecture:** docs/TECHNICAL_HANDOFF.md
- **MBA Presentation Guide:** docs/MBA_GUIDE.md